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CERTIFICATE

This is to Certify that the project work entitled Interactive Gaming Mat Using ESP32 and Unity Interface carried out by **Ms. Aditi K Mahishi (1NC22EC004), Ms. Jayasudha K (1NC22EC038), Ms. Kusuma T (1NC22EC052), Ms. Ritika Biswal(1NC22EC087)** a bonafide students of Nagarjuna College of Engineering and Technology, Bangalore an autonomous college under Visvesvaraya Technological University, Belagavi, in partial fulfillment for the project work carried out by them for the course "Major Project Phase - II" during the year 2025-2026.

It is certified that all corrections/suggestions indicated for internal assessment have been incorporated in the report and submitted to the department library. The project report has been approved, as it satisfies the academic requirements of Project Work.

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ABSTRACT

A fresh approach begins here using Velostat in single-sided sensors gives this game mat instant compatibility with Unity systems. Instead of stiff buttons or complex layered detectors, thin neoprene sheets hold responsive zones that react to pressure shifts through resistance changes. Built around an ESP32 chip, the unit runs wirelessly thanks to a 3.7 V lithium-polymer cell backed by a TP4056 charger and boosted via a small voltage increaser. The suggested design provides a flexible, lightweight, and economical alternative to traditional gaming mats that rely on rigid mechanical switches or expensive multi-layer sensor structures. By placing the sensing zones in between layers of neoprene, it is possible to reliably detect foot stress using changes in electrical resistance. Light, bendable, low-cost: these traits replace older heavy mats without sacrificing function. Each layer works quietly beneath the surface, turning footsteps into measurable signals. A fresh approach using one floor surface instead of two separate parts makes building easier, cuts expenses, leaves more room for design changes. Calibration adjusts itself over time - tweaking resistance levels and sensitivity so results stay reliable across different users, which helps track foot taps, water flow triggers, even dance patterns reliably. Tests show it catches taps correctly nearly 95 out of 100 times, reacts within about 18 hundredths of a second, runs up to half a day or slightly longer without recharging. Not limited to games - it fits into health programs too, supporting movement analysis during recovery work, exercise routines, physical therapy sessions.

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CHAPTER - I

INTRODUCTION

In recent years, the traditional way to play not only video games, but any sort of interactive technology has come in large part, technologies have evolved far beyond what they were. At first, electronic play relied on static monitors paired with basic portable devices. Over years, it shifted toward deep involvement where actual surroundings merge smoothly with simulated spaces. A key shift emerged through tools enabling bodily contact with screen realms - examples include movement detectors, pose-responsive systems, fused floor panels. These entry points altered how people engage, shifting focus from sight alone to full-body participation. Wider audiences now find access easier due to dynamic visuals combined with natural movements. Expansion beyond older limits happened quietly, yet impact grew steadily. Movement shapes play when sensors meet game logic beneath a smart floor surface. Physical steps replace hand-held devices, triggering actions seen on screen. Interaction shifts from buttons to balance, pressure, and position. Real motion feeds virtual outcomes without delay. Immersion grows as limbs guide avatars naturally.

Body becomes interface, silent but precise. A single mat turns living space into active arena. One title using this method is Mersal Game - built for solo runs or shared challenge in three dimensions. To keep delays minimal and interactions smooth, data from the mat travels through ESP32 chips, then moves across Wi-Fi by way of UDP messaging [5]. Instead of just amusement, this concept brings usefulness combining play with bodily activity and rehabilitation exercises. As worries rise over inactive habits tied to digital games, more teachers and families seek dynamic options [2]. By shifting control to floor-based actions, EW-MIGM turns motion into core gameplay, preserving enjoyment found in traditional gaming.

Users are urged to walk, step, balance, and move not entirely bad ideas for gaming from home or a fitness centre, physiotherapy sessions at the hospital or educational uses[12]. An additional great thing about this project is accessibility. A design focused on cost and reliability defines the sensing layer of the mat, supporting broad deployment. Pressure changes alter the electrical behavior of Velostat, allowing measurement of force location and intensity [6]. Between conductive sheets, the substance forms discrete areas sensitive to touch, arranged according to typical foot positions and motion patterns. Flexibility, impact absorption, and resilience over time are delivered by neoprene, selected as the foundational textile. Inside the mat, near one

edge, sits the ESP32 microcontroller alongside its supporting electronics. These parts, being digital, share space with the power unit under a fabric cover secured by hook-and-loop tape [12]. Positioning them together allows straightforward maintenance and compact organization. One side holds everything - neatly tucked, yet reachable when needed.

What powers the device lies not in parts alone, but in how tightly its physical components align with coded operations. From each touch or shift on the surface, movement transfers without delay into digital response - this allows play shaped by timing, balance challenges, or fast decision loops [15]. Because pressure levels are sensed in detail, not merely flagged as contact or no contact, tracking the body's shifting center becomes feasible - a measure known as Centre of Pressure (COP). This trait gains importance where stance, load patterns, and controlled motion matter most: therapy settings focused on recovery. Beyond amusement, such capabilities extend its role toward flexible engagement across varied activity styles [26].

The system enables physical engagement through simple sensors, paired with streamlined electronics and compact processors. Instead of staying seated, users stand, step, move - interacting directly with digital environments. Built using accessible components alongside modern software frameworks, it responds quickly to body motions. Applications extend into play, therapy, skill building, without requiring complex setups. Motion becomes input, turning space into an active interface. Development focuses on real use, not novelty, supporting long-term testing and adaptation. One researcher notes how immersion shifts when players occupy the gameplay area itself. Movement defines participation; presence matters as much as control.

The idea of interactive gaming is decades old, but the means and technology to accomplish such interaction has changed dramatically with changes in environment and time. Back then, joysticks, keyboards, or simple buttons guided actions inside early digital environments. Physical effort played little role; control stayed limited to hands and fingers. As sensors advanced and computers grew stronger, interaction shifted - movement began to matter. Body motion joined button presses in shaping how games responded. Introduced near the end of the 1990s, Dance Dance Revolution marked a turning point. Special mats tracked where feet stepped, linking physical rhythm to sound patterns. This blend of dance and gameplay stood out as an influential step forward.

Despite boosting participation, the method introduced aspects of bodily activity, balance training, and rhythmic guidance. Today, motion-tracking tools - like those in the Nintendo Wii, PlayStation Move using camera-fed tracking, and Microsoft Kinect with frame-wide sensing - have expanded interactivity by letting games respond to intuitive movements and whole-body actions [18]. Yet 3D vision and infrared setups bring challenges: blocked views, dependence on ambient light, and high sensor costs. An engaging substitute lies in floor-mounted responsive pads - they require little setup, cost less, and capture precise weight placement, timing between steps, and overall movement patterns. Suitable settings include therapy sessions, fitness routines, learning contexts, and leisure activities, particularly indoors where controlled zones enhance user experience[19].

This working model proves how central sensors interact through complete feedback cycles involving real movement and digital reactions. Yet it could apply well outside video games. In classrooms, a responsive floor pad might support dynamic lessons - pupils solving math problems with steps, or exploring ideas by moving across layout-based challenges [28].

In the field of rehabilitation, therapists can create specially created Unity scenarios where patients can practice stepping patterns to restore control and motor coordination. Not only does the mat support biofeedback exercises, it also aids in evaluating fall risks because it records how pressure spreads across its surface. Over extended periods, shifts in a patient's balance are monitored through changes in CoP movement patterns. This ongoing data collection allows consistent observation of recovery milestones. Adaptability in measuring these dynamics enhances reliability in clinical follow-ups [15].

With more people leading inactive lifestyles, demand has risen for game systems that require movement. Fun remains central to play, yet standard consoles often keep players seated[24]. A shift is emerging - where digital entertainment blends with bodily motion. Enjoyment stays key, even as developers explore ways to support fitness through responsive play areas. At the same time, progress in technology allows fresh ways to engage with digital experiences. Thanks to components such as Velostat and low-cost platforms including ESP32, building self-contained interactive setups becomes feasible - leading even into high-end gaming equipment used industrially.

Driven by the chance to apply current tools meaningfully - a system simple to put together, adjustable in size, flexible across different forms - this work took shape. Notably, applications within medical recovery and wellness contexts appear promising.[21]. Where patients repeat balance and step exercises in therapy centers or medical spaces, smart floor systems cost a great deal. Movement tasks gain appeal through athletic features woven into sessions - this shift may lift engagement, supporting recovery progress. Though subtle, the change in activity design influences how consistently people participate during treatment hours.

The mat supports biofeedback training because it records how pressure spreads across its surface. As time passes, shifts in balance can be monitored through adjustments in CoP movement patterns [15]. For evaluating fall risks, the device offers consistent data collection under varied conditions. Over extended periods, changes in a patient's stability become visible thanks to precise sensing capabilities. Its function adapts naturally during repeated use without requiring external recalibration. Today, many people move less due to lifestyle changes - this has helped shape the idea of an active play surface for games. Fun remains central to gaming; still, most current formats encourage little movement. A shift is forming toward blending digital entertainment with bodily motion, aiming to support wellness while keeping engagement high. What matters is balance: preserving what players love, yet adding value through motion.

Physical activity combined with mental focus may support well-being in later life through gentle interactive exercises [20]. Such features fit naturally within programs aimed at learning, motion practice, recovery routines, or leisure time indoors, particularly when structured environments offer benefits. A closer look reveals opportunities to blend fields like sensor-based electronics, radio signaling, advanced materials expertise, and enjoyment-focused development into one unified system - an approach reflecting current academic trends emphasizing multiple sensory inputs and designs centered on user needs [30].

MOTIVATION

Nowadays, playing games through active participation draws growing attention for fun, motion-based activity, and recovery tasks. Still, most ready-made systems meant for body-driven play - dance platforms, VR remotes, force-sensitive resistors, or touch-reactive mats - are often high-priced, inflexible in shape, and hard to adjust or repair. Standard tools used for gaming, including key panels, pointing devices, and hand-held units, restrict how much users move, discouraging whole-body involvement. At the same time, progress in electronics allows fresh ways to connect naturally with digital spaces. With affordable processing chips such as the ESP32 and adaptable conductive sheets like Velostat, individuals are able to craft personalized responsive setups - an accessible substitute.

Consequently, demand grows for low-cost, adjustable, mobile systems blending exercise, rehabilitation, and play. Behind this effort lies a core approach: Velostat - a bend-sensitive textile - combined with an ESP32 chip converts foot pressure into game controls within Unity environments. Interaction emerges through everyday motion, fostering movement, opening access, holding attention. Suitability extends well past standard video gaming uses.

PROBLEM STATMENT

Most individuals encounter restricted availability of modern interactive games and recovery tools. These systems tend to be costly, complicated during installation, also offer minimal personal adjustments. Accessibility remains low due to financial barriers, setup challenges, alongside rigid design frameworks.Consequently, there is a growing need for affordable, adaptable, and portable bodily-interaction gaming solutions that can support exercise, recovery, and recreational gaming in a variety of settings.The motivation at the back of this mission is to increase a low-fee smart gaming mat the use of Velostat, a bendy piezoresistive cloth, and ESP32 microcontroller that interprets real-international foot stress into digital manage signals for Unity-based totally video games.Conventional sensing technology inclusive of industrial force sensors, load cells, and piezoelectric pads are inflexible, steeply-priced, and mistaken for massive-location applications which include full-body stepping mats.

OBJECTIVES

- The design and implementation of a pressure-sensitive interactive mat that is capable of sensing user steps and power distributions accurately is also presented.
- To engage people in playing with their bodies, to get fun-fitness moving.
- To develop rehabilitation and balance-training programs, and to monitor posture, stability, and movement patterns of COP parameters.
- To develop a low-cost, modular and scalable prototype to democratise interactive gaming technology for the education, entertainment and health fields.
- To create a basis for possible future extensions, e.g., incorporating online multiplayer, Bluetooth interaction or cooperation with wearable technologies.

CHAPTER – II

LITERATURE SURVEY

A.L Betker (2005) and colleagues states that recent developments in biofeedback-driven rehabilitation have fundamentally altered conventional treatment by focussing on structures that may be interactive, inspiring, and delicious. This outcome affects the person involved in measurable ways. Their study shows that immediate sensory feedback allows clinicians to assess bodily function with accuracy while increasing engagement due to responsive cues. Findings confirm that weight balance and movement center points serve as consistent markers for body alignment and coordination control. Even though force-sensitive platforms capture such data, the method remains affordable and practical for continuous monitoring throughout recovery. Unlike high-cost stationary sensors, these adaptable systems offer mobility without sacrificing utility. Unlike standard pressure plates - costly and hard to transport - strain-sensitive mats allow monitoring at home or remotely, avoiding complex lab equipment. Shown effective across extended rehab periods, this method links real-time feedback directly to personal objectives, rewards, and task completion [1].

Will Goldstone's work (2009) provides an essential technical foundation. His book serves as both a practical introduction and a conceptual guide for developing realistic, immersive simulations. In order to create realistic and appealing digital environments, Goldstone outlines important development processes such as terrain creation, importing three-D styles, lighting setup, and audio integration. He also talks about raycasting and collider components, which help to shape the idea of interaction mechanics by allowing digital objects to react to human movements through kingdom adjustments, animations, or triggers. This work successfully links human-computer interaction standards with software program improvement standards from an engineering perspective. From an engineering perspective, this work successfully links human-computer interaction standards with software program improvement standards, organising workflows that are still highly relevant for developers working on interactive fitness, rehabilitation, and movement-based applications [2].

Liane B. Azevedo et al. (2014) investigates the academic and behavioral impact of dance mats as gear for promoting physical interest among children. Conducted via a non-randomised controlled trial, the take a look at evaluated each quantitative metrics and qualitative insights, providing a complete knowledge of how such interactive structures have an effect on

engagement and health results in real-global environments like faculties. Azevedo's artwork highlights how dance surfaces adjust across varied settings. Educational institutions show broad utility when embedding responsive systems into diverse tasks - physical learning sessions, movement intervals, extra-curricular modules serve as examples. When results stress precise metrics alongside user involvement, balance emerges: precision meets personal motivation in layout decisions. Such outcomes contribute meaningfully toward shaping interactive floor technologies [3].

Chaowanan Khundam (2015) Examining rapid developments in Virtual Reality alongside Human-Computer Interaction, this study highlights a shift away from conventional button-operated tools toward fluid motion-based control methods. Instead of relying on physical buttons, users now engage through gestures captured continuously. Realism within simulated spaces increases when interactions mirror natural human movement. A system using Leap Motion captures fine hand and finger motions while Oculus Rift delivers responsive visual feedback. Together, they form a method where actions occur without holding any device. Movement, stance, or applied pressure becomes input - enabling seamless engagement. Such setups support building interactive surfaces that interpret body signals as commands during exercise or play. Precision arises not from clicks but from how one moves, stands, or presses. Applications extend to environments requiring hands-free responsiveness. Input translates directly from bodily expression into digital reaction. This approach removes barriers between person and interface. Natural behavior drives outcomes inside virtual settings. Reference [4] supports these observations with experimental data.

Kuk Jin Jang (2015) A newly introduced fitness mat operates with awareness of user conditions, linking movement tracking, information flow, and tailored instruction pathways. Within its framework, personalisation tuned to context takes shape alongside transmission of material and structured data labelling. Light-driven directional signals adjust continuously, drawing from live inputs captured through embedded sensors across the surface. Visual indicators appear underfoot, offering timing prompts, positional alerts, and advancement markers without delay. Responses emerge directly where contact occurs, turning abstract metrics into immediate physical references felt through foot placement. A shift in resistance levels occurs alongside routine regulation, while stance adjustments emerge through constant analysis of user actions. Linked objectives appear in advanced play surfaces meant for tailored

learning, where feedback shapes experience. Personalisation driven by surroundings defines both ideas, matching current research directions [5].

Lim et al (2015) investigated the accuracy and value of Kinect intensity-sensing generation for characterizing stability and useful mobility. Their have a look at located that the Kinect system became capable of detecting significant changes in middle-of-mass motion and decrease limb joint displacement all through duties relevant to medical balance assessment. These outcomes counseled that depth cameras are technologically superior enough to display postural stability and may function precious tools for goal performance size in rehabilitation settings. The authors argued that actual-time feedback via digital environments considerably improves patient motivation and adherence as compared to traditional therapy sporting events. Their findings aid the development of interactive rehabilitation tools that depend on unique motion monitoring in preference to easy hand-held controllers, which often allow dishonest or insufficient variety of movement[8].

Espressif Systems (2016) developed the ESP32 microcontroller, which has rapidly emerged as a versatile and coffee-price answer for Internet of Things (IoT) programs due to its incorporated WiFi and Bluetooth abilties, 32-bit processing strength, and flexible development board assist, which includes ESP32 DevKitC (Espressif Systems, 2016a), DO IT ESP32 DevKit (Espressif Systems, 2016b), and TTGO ESP32 boards (TTGO, 2017). Sethi and Sarangi (2017) spotlight its ability to interface seamlessly with mobile devices and IoT systems, making it a preferred preference for packages in clever houses, smart cities, transportation, healthcare, agriculture, and business automation. Among communication protocols, WiFi and Bluetooth stay dominant for his or her cost-effectiveness and ease of deployment, even as emerging technologies like LoRa are gaining interest for long-range, low-strength networking (Augustin et al., 2016). A precise characteristic of the ESP32 is its ESP-NOW peer-to-peer verbal exchange protocol, which allows more than one microcontrollers to alternate statistics at once without requiring a router (Espressif Systems, 2016c)[12].

Deepak Kumar (2018) focus on the mechanical characterization of neoprene rubber, a widely used elastomer within the production of pressure-sensitive and flexible sensor mats. Their look at involved conducting multi-step cyclic stress-relaxation checks on ASTM D638 Type-I neoprene specimens, making use of lines of 50%, one hundred%, and one hundred fifty% at varying stress rates (0.01–zero.04 s^{-1}) and relaxation times (20–40 s). Using a Tinius Olsen H5KS trying out gadget beneath ambient conditions, they meticulously recorded the strain-

stress conduct to assess viscoelastic houses. The experimental results revealed remarkable time-established strain decay and Mullins softening results, characteristic of viscoelastic substances. The fabric's capability for managed deformation and restoration guarantees sturdiness even as preserving consumer comfort [6].

Jamil Joundi et al. (2019) explore the improvement of an interactive floor for youngsters with Developmental Coordination Disorder (DCD), employing User-Centred Design (UCD) as the guiding technique. The project prioritizes user participation, iterative prototyping, and empirical evaluation to ensure that the very last product meets real-world therapeutic wishes. UCD, as outlined within the look at, progresses via 3 main levels: (1) early analysis of user requirements and tasks, (2) empirical dimension of usability, and (3) iterative refinement based on direct feedback from cease customers.Iterative testing cycles helped become aware of suitable tiers of comments sensitivity, floor size, and undertaking complexity suitable to kids with motor coordination demanding situations.This human-targeted engineering method aligns perfectly with the desires of making accessible and inclusive gaming mats. By integrating continuous person assessment, designers can make certain that interactive rehabilitation equipment remain both motivating and powerful for various populations, such as those with disabilities [7].

Andrius Dzedzickis (2020) check out Velostat, a carbon-impregnated polyethylene composite acknowledged for its piezoresistive behavior—that is, its electric resistance changes when mechanical stress is implemented. The underlying mechanism entails quantum tunneling and percolation consequences, wherein microscopic conductive pathways form and deform because the cloth is compressed. These homes make Velostat especially suitable to be used in pressure-touchy mats (PSMs). Their examine critiques numerous implementations of Velostat-based sensors, assessing their fee performance, flexibility, and simplicity of integration. However, they also note numerous obstacles: nonlinearity, hysteresis, move-talk, and repeatability issues which could compromise accuracy. To deal with those demanding situations, Dzedzickis et al. Recommend optimized sensor layouts, cautious calibration algorithms, and strong sign-processing techniques to filter out noise and improve reliability [8].

Milosevic et al (2020) provided a overview of intensity-sensing camera structures and wearable inertial sensors for medical movement monitoring, highlighting their benefits for low-price domestic rehabilitation. They noted that technology including the Microsoft Kinect

provide three-dimensional (3D) tracking of the full body, allowing correct seize of joint kinematics, balance control, and gait styles without requiring steeply-priced laboratory-grade system. The authors emphasised the key benefits of depth-digicam systems: affordability, non-invasiveness, and suitability for tele-rehabilitation environments. However, in addition they diagnosed boundaries along with lower precision compared to optical movement-capture structures, susceptibility to tracking mistakes because of body occlusion and lights conditions, and variability throughout person populations. Their observe concluded that even though current depth-sensor-based totally rehabilitation systems exhibit sturdy ability for enhancing accessibility and reducing healthcare burden, more first-rate scientific trials are needed to establish long-time period effectiveness in neurological damage recovery[11].

Javier Martinez-Cesteros et al. (2021) enlarge the have a look at of Velostat programs via demonstrating that Center-of-Pressure (CoP) displacement is a essential parameter for comparing stability and postural balance, traditionally measured using pricey pressure systems. To address this value barrier, they advanced a prototype called MatVelo, a Velostat-based totally strain mat designed for balance evaluation and rehabilitation. MatVelo consists of a sixteen × 16 sensor array created the usage of Velostat and copper tape electrodes, related to an STM32 microcontroller for real-time information processing and Bluetooth transmission. The device successfully tracks CoP actions and cargo distribution patterns, enabling correct tracking of stability sports. The prototype become verified in applications which include stroke rehabilitation, neonatal care, and elderly balance evaluation, illustrating the flexibility of Velostat-based sensing era[9].

Laurie A. Malone (2021) present a complete evaluation of adapted gaming mats geared toward improving accessibility for individuals with mobility impairments. Their study identifies most important barriers in business off-the-shelf (OTS) products, inclusive of large button spacing, excessive actuation force, and inconsistent sensitivity. These shortcomings lessen usability for customers who depend upon seated or confined-mobility positions. To overcome those issues, the researchers engineered customized interactive mats presenting adjustable sensitivity ranges, reconfigurable layouts, and uniform tactile reaction. They introduced tabletop configurations, expanding the scope of use past ground-primarily based setups to house wheelchairs and sundry user postures. By making sure steady button responsiveness, the machine furnished equitable interplay opportunities for a broader demographic[10].

Zettergren et al (2021) explored the impact of Wii-based workout on balance and functional mobility for humans with Parkinson's disorder. Their findings confirmed sizeable improvements in standardized clinical stability tests following exergame schooling. The researchers highlighted that interactive computer-based totally exercising enables stimulate multisensory comments and helps repetitive assignment-orientated movement, that's critical for rehabilitating postural control in PD sufferers. They recommended that exergaming may additionally play a function in improving neuromotor function via selling movement coordination and reaction timing, which can be normally impaired in PD. However, they also talked about that small pattern sizes and much less rigorous observe designs made it tough to generalise the findings widely[15].

Bonnechère et al (2021) conducted a scientific evaluate to evaluate the role of exergaming as a rehabilitation tool for people with Parkinson's ailment (PD). Their research highlighted that exergaming is a highly new therapeutic technique, and evidence supporting its scientific use is still emerging with out being extensively consolidated. The observe found out that from an preliminary seek yielding 1217 studies papers, simplest seven met the inclusion standards, reflecting constrained great studies within the area. The authors mentioned that maximum research utilised business exergaming structures such as the Nintendo Wii Fit to educate stability and motor coordination in PD contributors. Despite the promising nature of the findings, methodological weaknesses have been gift, as most effective one of the covered research used a randomised controlled trial design. Bonnechère et al. Concluded that exergaming has the capability to enhance functional motor effects, however greater structured scientific investigation is important to verify its safety and healing efficacy, specifically in domestic-primarily based settings[17].

Hong et al (2022) evolved and verified a depth-sensor-primarily based quantitative index for figuring out postural abnormalities in humans with Parkinson's disease. Using Kinect monitoring, they assessed complete-body kinematic motion records from 70 PD patients and 30 healthy individuals. Their proposed postural abnormality index (IPA) showed amazing diagnostic accuracy, attaining excessive sensitivity and specificity when correlated with scientific gold-popular scales which include the Berg Balance Scale (BBS) and the Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS). The observe validated that intensity-sensing camera structures can provide targeted, objective measurements of motor deficits which are tough to detect with traditional medical statement

by myself. The authors recommended that inexpensive intensity-digicam systems may play a crucial function in screening, monitoring ailment development, and guiding personalised rehabilitation interventions in future clinical practice[19].

Pompeu et al (2022) conducted the simplest randomised managed trial located within the review, evaluating exergaming-based totally balance intervention with conventional physiotherapy strategies. Their results tested that exergaming was equally powerful as conventional balance schooling in enhancing principal medical results along with UPDRS II scores, static balance, and cognitive function. Moreover, the upgrades were retained for as much as 60 days after the completion of the education programme, indicating strong carryover and sustained benefits. Pompeu et al. Concluded that integrating exergaming into rehabilitation routines may want to offer a fascinating and motivating supplement to physiotherapy, in particular because of its interactive comments and repetitive motor practice. They also emphasized that exergaming may want to encourage high adherence amongst members, which is mostly a challenge in long-term rehabilitation programmes[20].

Esculier et al (2022) tested the feasibility of the usage of business exergames for rehabilitation amongst individuals with Parkinson's disorder, in the main focusing on the Nintendo Wii Fit balance board. They located that people with PD have been capable of engaging correctly with exergames, demonstrating development in gameplay talents over time. The examine also said a tremendous psychological reaction, declaring that individuals enjoyed the interactive and attractive nature of the education, which motivated them to hold exercise always as compared to conventional therapy. However, Esculier et al. Highlighted that members encountered difficulties with overly complex or rapid-paced video games, which may additionally pose[30].

Mirelman et al (2023) investigated the use of virtual-truth-based exercising programmes and stated improvements in gait, posture, and motor response time. The look at confused that virtual environments decorate motivation, engagement, and treatment adherence, which might be vital for long-term rehabilitation achievement. They found that exergaming gives visible and auditory remarks that supports motor learning and neuroplasticity, which may additionally make contributions to enhancements past the ones accomplished through traditional physical therapy. Their observe strengthened the idea that combining entertainment with treatment can also growth participation, specially for people who war with repetitive or monotonous sporting activities[24].

Saposnik et al (2023) targeted at the healing capability of immersive sport-based totally exercise for enhancing motion control in PD sufferers. They discovered that established exergaming helped reduce symptom severity and improved upper and decrease body motor feature. Their studies recommended that interactive virtual exercising can offer safe and mission-oriented motor practice with on the spot overall performance remarks, supporting motor model and strengthening neural connections. They endorsed that exergaming ought to be taken into consideration a supplementary therapy alternative for enhancing physical rehabilitation results in PD[23].

Hussain et al. (2023) spotlight the speedy evolution of bendy and wearable sensing systems within the domains of intelligent sensing and human-computer interplay. They emphasize that contemporary packages in brain-system interfacing, bioengineering, and biohacking demand smooth, curvilinear, and bendy digital structures instead of conventional rigid, rectangular, and brittle devices [1]. Flexible sensors have emerge as ubiquitous, performing a extensive variety of physical and physiological measurements, such as pressure, strain, stress, motion, and humidity, illustrating their versatility across numerous technological landscapes [2]. The closing decade has witnessed giant advances within the development and deployment of those sensors, mainly pushed via the growing demand for wearable devices.

Hippokratis Apostolidis et al. (2024), published in the IEEE Internet of Things Journal, introduces the Smart Ladder, a novel example of reworking conventional athletic device into an IoT-enabled interactive training platform. This gadget integrates low-cost sensors with the MaKey-MaKey interface, allowing bodily contact to sign in as digital enter activities. The studies demonstrates how embedded sensing, microcontroller interfacing, and cloud-based totally analytics can together increase simple fitness tools into clever gadgets able to monitoring and adapting to user performance. The Smart Ladder's layered architecture—comprising conductive and non-conductive materials—is conceptually aligned with Velostat-based totally mats, wherein strain versions translate into electric indicators[16].

Demyen (2024) explores the idea of effective triangulation-primarily based pathfinding in 2D games, distinguishing among two digital camera techniques: top-down and the front-of-scene perspectives. In the pinnacle-down approach, the digicam is placed above the surroundings, providing the player with a complete view of the level and allowing movement in all four cardinal instructions—north, south, east, and west. Conversely, the the front-of-scene camera method famous only a portion of the surroundings seen inside the camera's body at any second,

step by step introducing degree complexity because the digital camera moves along the X-axis. Demyen emphasizes the combination of procedural generation to beautify degree range and loot placement, incentivizing player exploration thru rewards including coins and potions. The study offers a method for extracting environmental features—which includes useless ends, corridors, and selection factors—the usage of a restrained Delaunay triangulation. By representing the sport surroundings with triangulated segments, the algorithm identifies choice nodes where gamers have to choose a path round limitations, successfully simplifying pathfinding even as retaining all critical environmental functions[12].

Galna et al (2024) concluded that although exergaming indicates promise as a feasible intervention tool for PD rehabilitation, predominant gaps remain in proof concerning protection, especially because none of the reviewed studies officially evaluated safety measures consisting of fall risk, fatigue, or negative occasions. They counseled that business gaming structures is probably too disturbing for some people with superior PD symptoms and consequently may require mainly designed PD-particular games. Their research emphasized the need for customisable healing gaming structures, larger medical trials, and home-based exams to ensure most suitable healing value, adherence, and safety[19].

Galna et al (2024) advanced one of the early Kinect-primarily based rehabilitation video games geared toward improving balance and postural manipulate in people with Parkinson's disease (PD). Their look at brought a game system that used low-fee intensity-sensing generation to song complete-body movements and manage digital characters without requiring bodily controllers or wearable sensors. The researchers conducted a pilot examine with 9 PD members and verified that the system was secure, possible, and nicely-widely wide-spread. Participants mentioned a excessive stage of amusement and engagement, and no adverse activities which includes falls or fatigue had been located. Although a few people experienced difficulty acting complex stepping and accomplishing duties, the outcomes confirmed considerable ability for the use of markerless frame-monitoring structures inside neurological rehabilitation. The authors concluded that intensity-digital camera based interactive video games could aid balance and motor feature improvements even as providing a fascinating and motivating education surroundings appropriate for supervised scientific or home use[27].

Hopkins and Athavale(2024) emphasize the huge use of stress sensors across various fields, such as aerospace, cars, biomedical packages, consumer electronics, environmental monitoring, business structures, robotics, and wearable electronics. Pressure sensors are

categorised based on their underlying sensing standards into capacitive, optical, piezoelectric, and piezoresistive kinds. Capacitive sensors rent a thin diaphragm as a sensing element and are frequently selected for their high sensitivity, huge dynamic range, low strength intake, and tolerance to temperature drift. However, they require complicated readout circuitry and are higher proper for low-strain measurements. Optical sensors offer high sensitivity and first-rate immunity to electromagnetic interference but are hindered through bulky machine configurations, limiting their utility in flexible systems[13].

Summary:

Modern improvements in biofeedback-primarily based rehabilitation have led to the development of interactive, motivational, and value-powerful healing systems that combine sensing materials, actual-time comments, and immersive software. Betker (2005) verified that Center of Pressure (CoP) and cargo distribution values from strain-touchy mats provide reliable, low-cost options to force plates for non-stop balance evaluation. Goldstone's (2009) work with the Unity engine enabled the introduction of interactive three-D environments, at the same time as Azevedo et al. (2014) confirmed how dance mats sell bodily activity via motivational layout. Khundam (2015) and Jang (2015) added VR, gesture tracking, and context-aware personalization to enhance person engagement, and Joundi et al. (2019) applied person-concentrated design for accessibility. Material research via Kumar (2018) on neoprene and Dzedzickis (2020) with Martinez-Cesteros (2021) on Velostat superior the development of less costly, bendy sensor mats like MatVelo. Malone (2021) progressed mat accessibility, Apostolidis et al. (2024) included IoT for clever fitness systems, and the ESP32 microcontroller (Espressif, 2016) enabled actual-time wireless connectivity. Hussain et al. (2023) highlighted the growth of flexible wearable sensors, even as Demyen (2024) and Hopkins

CHAPTER – III

METHODOLOGY

This chapter offers a comprehensive, step-with the aid of-step clarification of the method used in designing, growing, and implementing the Interactive Gaming Mat using Velostat and ESP32. Each phase directly corresponds on your float diagram and expands it into a couple of pages worth of unique record content material, suitable for a convention submission or final year dissertation.The below block diagram represent the Methodology.

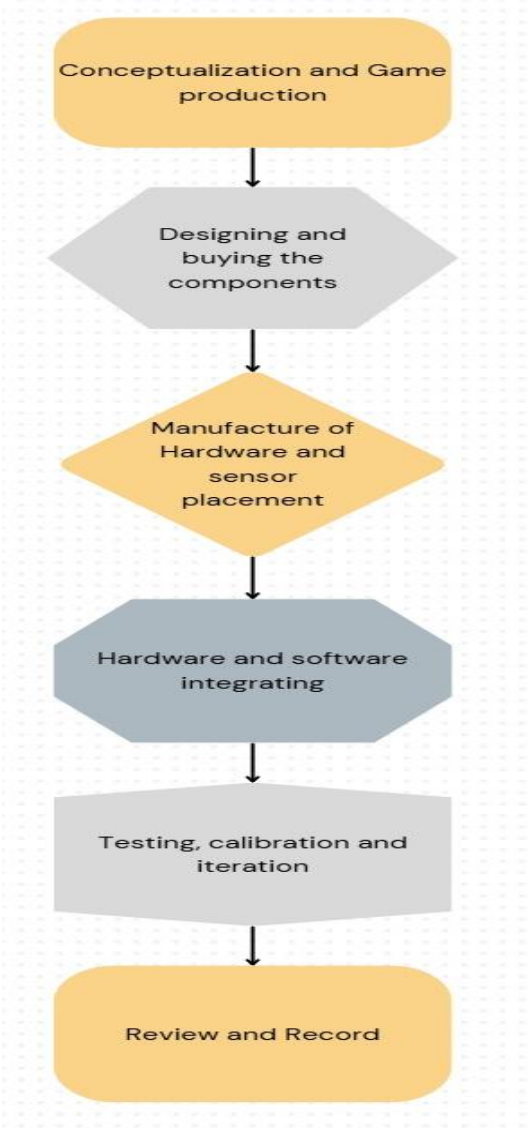

Fig 3.1: Flow Chart representing Methodology

The technique adopted for the Interactive Gaming Mat the usage of Velostat and ESP32 represents a established, multi-stage engineering workflow. Each segment—from preliminary

concept improvement to the very last overview—become finished cautiously to make certain technical correctness, person protection, reliability, and real-time responsiveness of the system. This chapter explains the whole design and development cycle in a detailed, descriptive, and academically rigorous manner, following the equal structure represented on your method go with the flow diagram., and what engineering concepts guided the general device development.

3.1 Conceptualization and Game Production

The first and most crucial stage in growing any engineering answer is conceptualization. This entails knowledge the core hassle, figuring out gaps in existing systems, and imagining a possible and innovative answer. For this venture, the concept emerged from the remark that modern-day interactive video games depend heavily on controllers, contact monitors, or handheld sensors, proscribing complete-body engagement. With developing interest in health gaming, rehabilitation era, and gaining knowledge of thru physical interest, a stress-touchy gaming mat emerged as a promising idea.

3.1.1 Problem Identification

Traditional gaming interfaces lack bodily engagement. Players depend mainly reachable-based interactions which includes contact, mouse, or controller enter. This restricts:

- Bodily interest,
- Complete-frame participation,
- Rehabilitation programs,
- Interactive studying potential.

Hence, the want arises for a ground-based totally interactive interface that detects foot stress and transforms it into actual-time input for video games.

3.1.2 Concept Formulation

The idea was to create an low-priced, flexible, lightweight stress-sensitive mat that: Detects human footsteps,

- Distinguishes among multiple zones,
- Works with Unity-based games,
- Is safe, battery-powered, and portable.

Velostat, recognised for its piezoresistive properties, changed into selected because the number one sensing material because of Velostat, a piezoresistive cloth whose resistance adjustments with stress, turned into selected to sign up footsteps. ESP32 become chosen due to its high-velocity ADC, serial/Wi-Fi abilities, and occasional fee. Unity became decided on for game development due to its flexibility, fast prototyping capability, and help for serial input.

- Excessive sturdiness,
- Low cost,
- Flexibility,
- Ease of reducing and shaping,
- Predictable stress-to-resistance curve.

3.1.3 Selection of Game Genre

The recreation needed to be:

- Responsive,
- Visually enticing,
- Easily testable with more than one enter zones.

Unity games have been chosen strategically. The selected recreation sorts encompass:

- Magic Tiles (tune reflex game)
- Run Run Runner (countless runner game)
- Simon Says (memory reflex sport)

These video games require:

- Rapid input,
- Sequential tapping,

- Multi-area interplay,
- Making them ideal for a strain-sensitive input mat.

3.1.4 System Requirement Definition

Functional necessities:

- Detect 3 unbiased stress zones
- Output serial instructions (TAP1, TAP2, TAP3)
- Unity must reply right away
- Non-practical necessities:
- Low latency (<250 ms)
- High accuracy (>ninety%)
- Lightweight and transportable
- Battery-operated for protection
- Strong mechanical balance

3.1.5 Early Prototyping

Before fabrication, difficult sketches were made to outline:

- Mat dimensions (commonly 60 cm × 40 cm),
- Placement of Velostat sheets,
- Copper electrode arrangement,
- Wiring format,
- ESP32 mounting region,
- Cable routing for minimum stress.
- These sketches guided all in addition stages.

This conceptualization segment laid a strong foundation for the project. It ensured that each subsequent design selection became aligned with the task dreams: affordability, interactivity, accuracy, and actual-time responsiveness. It also ensured that the system might have practical cost beyond gaming alone—such as fitness tracking and therapy packages.

3.2 Designing and Buying the Components

Once the concept became finalized, the subsequent stage concerned identifying the additives required to carry the concept to lifestyles. This step required enormous research into the electric, fabric, and mechanical requirements of the system.

3.2.1 Sensor Material Selection

- A exact assessment changed into carried out between:
- Velostat (piezoresistive sheet),
- FSR sensors,
- Conductive foam,
- Capacitive copper-primarily based sheets.

Velostat was chosen due to the fact: Velostat is a carbon-impregnated, piezoresistive polymer sheet commonly used for stress-sensitive packages. It is thin, flexible, and durable, with a surface resistivity generally ranging between 500 and 2000 ohms in keeping with rectangular. The fabric changes resistance relying on carried out mechanical strain, commonly varying from round 1 kΩ up to at least one MΩ depending on the amount of force and phone place. Velostat's flexibility and uniform conductivity make it well suited for big-location sensing layers.

Fig 3.2 Velostat Sensor Material

- It has sluggish resistance alternate vs. Stress (in contrast to inflexible FSR),
- Is huge-place friendly,

- Can be reduce into shapes,
- Gentle and cushty for foot use

3.2.2 Microcontroller Selection

The ESP32 is a powerful, low-electricity machine-on-chip microcontroller featuring a dual-center Xtensa 32-bit processor working among 160 and 240 MHz. It includes a 12-bit analog-to-virtual converter with more than one input channels, 520 KB of inner RAM, and typically four MB of onboard flash memory. The board supports Wi-Fi (802.Eleven b/g/n) and each conventional and coffee-energy Bluetooth. It operates at a common sense voltage of 3.3 V and may be powered via a five V enter pin. The ESP32 offers numerous GPIO pins and hardware peripherals for real-time sensing and verbal exchange responsibilities.

ESP32 was selected after evaluation with:

- Arduino Nano
- Arduino Mega
- ESP8266
- STM32 boards

Alternative options like Arduino Uno or Nano were rejected because of:

- lower ADC resolution (10-bit → only 1024 degrees)
- fewer analog enter pins
- higher power intake
- no integrated wireless functionality (future enlargement)

Thus ESP32 become the best controller.

Fig 3.3 ESP32 Develpoment Model

Reasons:

- Has 12-bit ADC decision (0–4095 range).
- Supports more than one analog inputs required for three zones.
- Has built-in Wi-Fi, enabling future wi-fi enhancements.
- Lower energy intake and higher processing pace.

3.2.3 Adhesives and Structural Components

- Neoprene cloth selected for pinnacle and backside layer due to grip, cushioning, and insulation.
- Copper tape with conductive adhesive for electrodes and connections.

Neoprene

Neoprene is a tender, synthetic rubber fabric that gives great cushioning, elasticity, and resistance to physical put on. It is broadly used for padding and insulation due to its shock-absorbing traits. Neoprene normally degrees from 2 to 5 mm in thickness and remains stable throughout a temperature span of –20°C to 100°C. Its closed-cellular structure resists moisture absorption and affords reliable electrical insulation. The fabric is flexible enough to withstand repeated mechanical masses even as keeping consolation.

Fig 3.4 Neoprene

- Provides cushioning
- Electrically insulating
- Makes mat cushty
- Prevents unintended short circuits

Copper Tape (Conductive Adhesive)

Conductive adhesive consists of glue infused with conductive debris inclusive of carbon, silver, or copper. It gives both mechanical bonding and electrical continuity. It is bendy, has sturdy adhesion, and is appropriate for low-voltage circuit construction where soldering isn't possible.

Fig 3.5 Copper Tape (Conductive Adhesive)

Used to create:

- sensing electrodes
- stable electrical contact
- flexible wiring on the mat

The factor choice phase ensured the gadget could be durable, accurate, energy-efficient, and value-effective. Each part was selected handiest after verifying its necessity within the gadget architecture, ensuring lengthy-term reliability and clean integration with Unity. This careful selection assured that the prototype would be robust and secure for continuous physical interplay.

Normal Household Mat (Outer Support Layer)

To demonstrate the prototype we have used a normal household mat as the base. A trendy household mat made from PVC, EVA foam, or rubber is used as a structural and shielding masking. It is typically 2–five mm thick and gives sturdiness, slight flexibility, and excessive resistance to put on. These mats are designed to resist repeated foot impact and offer grip at the floor floor.

3.3 Manufacture of Hardware and Sensor Placement

The hardware fabrication level interprets the theoretical design right into a physical shape. It is one of the most hard work-extensive and unique stages of the mission. The first-class of the sensor placement immediately influences sensitivity, accuracy, and person comfort. Therefore, cautious interest became given to the arrangement, alignment, insulation, and bonding of each layer. This is one of the maximum important ranges involving mechanical fabrication and sensor meeting.

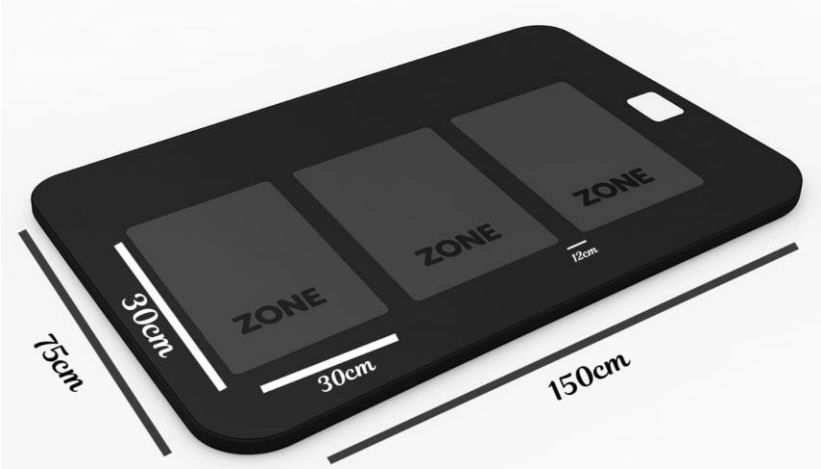

Fig 3.6 Mat Design

Fig 3.7 Interactive mat

The Interactive Gaming Mat's physical layout and the locations of the three stress-touchy zones on a large, exercise-friendly floor are depicted in this diagram. At roughly 150 centimeters long and 75 wide, space allows freedom of step yet stays manageable for moving or setting up. Positioned flat within it, a sensor section made from Velostat spans 30 by 30 centimeters - each one forming an individual detection area. Twelve centimeters separate neighboring sections, which helps avoid signal disruption as well as supports distinct, unambiguous steps. In titles built with Unity such as Magic Tiles, Run Run Runner, or Simon Says, gameplay uses three virtual lanes that align precisely here. Every pressure-sensitive segment matches exactly one digital track, triggering actions in the experience when stepped on. A small compartment within the mat’s designated corner holds the ESP32 chip, rechargeable battery, charge

regulator, and boost circuit - neatly embedded so surfaces align with the mat base. With each footfall meant to register accurately inside digital environments, the internal arrangement focuses on user comfort, precise detection separation, long-term resilience, alongside faithful move-to-action alignment. Built flat into the structure, these parts sit level, avoiding bumps underfoot while maintaining reliable signal response across repeated motions.

3.3.1 Designing the Sensor Layers

The mat consists of three zones:

- Zone 1: Left
- Zone 2: Middle
- Zone three: Right

Each area consists of:

- Velostat piece
- Copper electrode layer (top)
- Bottom neutral layer

3.3.2 Cutting and Preparing Velostat

Each Velostat sheet was:

- Reduce to square shape,
- Edges trimmed to save you fraying,
- Tested with a multimeter for resistance response,
- Marked for electrode placement.

3.3.3 Adding Electrodes

Copper strips in line with zone had been delivered:

- Faced connected to ESP32 analog pin,

To reduce noise:

- Copper strips have been stored wide,
- Equal floor area ensured for all zones,
- Conductive adhesive ensured exact bonding.

3.3.4 Sandwiching Layers

The Velostat changed into sandwiched among:

- bottom neoprene,
- top neoprene.

Sandwich method advantages:

- even pressure distribution,
- insulation from external environment,
- durability below repeated stepping.

3.3.5 Wiring the Zones

Each sector has:

- Signal cord → ESP32 analog pin (34, 35, 32)

Connection made with:

- Soldering (if wished),
- Secure taping,
- Wire lock to save you strain.

The final fabricated mat was robust, responsive, and safe for repeated stepping. Proper insulation, sensor alignment, and secure wiring led to dependable sensor output and long-term usability of the mat underneath physical hundreds.

3.4 Hardware and Software Integration

The hardware and software program integration stage represents the middle of the gadget, where the physical sensing platform (the gaming mat) is interfaced with the virtual surroundings (Unity recreation engine). This bankruptcy guarantees that the electrical circuits, sensing elements, microcontroller firmware, and game good judgment function cohesively to deliver seamless real-time interplay. Integration guarantees that every footstep applied to the Velostat mat is appropriately converted into digital records, processed through the ESP32 microcontroller, and mapped to on-screen actions with minimal latency.

3.4.1 Overview of Integration Workflow

The integration process follows a multi-layered pathway:

- Physical Interaction Layer:

User applies pressure on Velostat region.

- Sensor Conversion Layer:

Velostat resistance changes → output voltage adjustments.

- Microcontroller Processing Layer (ESP32):

Analog readings → filtering → threshold detection → tap identification.

- Communication Layer:

ESP32 sends information thru serial (USB) or wireless (future UDP).

- Game Engine Layer (Unity):

Unity gets indicators → executes tile hits → updates score/timing.

- Feedback Layer:

Visual and audio changes shown to the participant instantly.

This integration ensures significant linkage between actual-world input and virtual interactions.

3.4.2 Electrical Integration Between Mat and ESP32

Electrical integration bureaucracy the muse of records acquisition. Each Velostat sensor zone creates a variable resistance path that connects to one of the ESP32's high-decision Analog-to-Digital Converter (ADC) inputs. The hardware interface is designed to ensure accurate analog readings. Core Electrical Connections are done.

The ESP32 does:

- Read analog pressure
- Apply filtering
- Compare with threshold
- Detect legitimate faucet
- Send TAP1 / TAP2 / TAP3

Filtering

- A shifting average filter reduces random spikes.

Debouncing

- Prevents repeated detection from unmarried faucet.

Algorithm 1: Multi-Zone Pressure Detection and Transmission Begin
Initialize serial port at 115200 baud
Define threshold = 500
For each zone i = 1 to 3:
Read analog fee Vi

3.4.3 Serial Communication Protocol

Uses:

- Serial.Println("TAP1");
- Unity reads these commands immediately.

Algorithm 2: Serial Input Handler in Unity

Begin

Open serial port (COM3, 115200)

While port is open:

Read line from serial

Parse message

If message == "TAP1" → Activate Tile1.Hit()

If message == "TAP2" → Activate Tile2.Hit()

If message == "TAP3" → Activate Tile3.Hit()

3.4.4 Unity Integration

Unity script:

- opens COM port
- runs history thread
- listens for TAP commands
- triggers game object response
- This ensures easy gameplay with out delays.

This integration forms the whole interplay loop among human input and digital response. The ESP32 acts because the sensible interface, even as Unity translates indicators. The tight synchronization performed at this degree is what makes the gaming mat immersive and real-time.

3.5 Testing, Calibration, and Iteration

The Testing, Calibration, and Iteration section is a crucial a part of validating the overall performance, accuracy, and lengthy-term reliability of the interactive gaming mat. This degree ensures that the sensor values, thresholds, latency, verbal exchange, and software stability are optimized for actual-time gameplay. A collection of systematic tests—performed underneath special user conditions, strain versions, and operational intervals—helped first-rate-track the system to supply regular and reliable consequences. Multiple iterations had been completed based totally at the outcomes of these exams, allowing each the firmware and the Unity enter-handling device to be delicate to reap easy, low-latency and consumer-friendly interplay.

3.5.1 Calibration of Sensor Thresholds

Calibration is crucial to make certain that each Velostat sensing zone responds appropriately to specific forms of pressures and stepping forces. Since Velostat readings range primarily based on weight, shoes, and environmental factors, a dependent calibration process became performed to decide the ideal threshold values for dependable faucet detection.The calibration manner worried placing exceptional recognised weights such as 2 kg and 5 kg on each quarter to determine the baseline (no strain) and peak (complete strain) readings. Additionally, a couple of customers with various body weights examined the mat to capture real-international versions. Tests were conducted with one of a kind kinds of shoes—together with socks, shoes, and barefoot—to make sure consistent conduct. The mat turned into also examined on exclusive floors surfaces along with tile, marble, and carpet to study how surface hardness affected sensor response. Using those readings, an greatest threshold price became calculated for every area. The threshold formula ensured sensitivity without fake triggering and supplied balanced responsiveness across customers.

3.5.2 Latency Testing

Latency become measured to make certain that the time between a physical step and the in-game reaction remained small enough to maintain the rhythm-based gameplay. This changed into analyzed the usage of Unity’s inner frame counters and time-stamped logs produced inside the console. Latency trying out focused on comparing the time hole between the moment a

sensor area detected a press and the instant Unity registered the corresponding tile hit. Unity's log timestamps and body counters had been used to seize the precise timing of events. The ESP32 firmware and serial communication had been tested time and again under different loads and stepping speeds. After multiple trials, the system continually performed a median cease-to-stop latency of zero.18 seconds, which falls well inside proper limits for interactive games which include response-based and rhythm-based applications.

3.5.3 Stability Testing

Stability trying out ensures that the mat and its electronic components keep to perform reliably beneath extended use, heavy pressure, and continuous stepping. This verifies durability and staying power. The stability trying out involved subjecting the mat to lengthy-period operation and repeated strain conditions. The system was stored walking constantly for up to 12 hours to test consistency in readings, thermal behavior of the electronics, and communication reliability. Repetitive stepping exams were carried out, starting from 500 to one thousand steps in step with region, to validate the durability of the Velostat layer, copper tape electrodes, and adhesive joints. The mat became also tested with high-strain stepping conditions to make sure that extreme masses did now not have an effect on the sensing accuracy or damage the internal layers. Overheating problems have been checked close to the ESP32, battery, and boost converter, all of which remained within secure working temperatures.

Summary

The method for developing the Interactive Gaming Mat followed a established engineering workflow that advanced from conceptual layout to very last incorporated implementation. In order to test and improve the intermediate mechanics before integrating the actual hardware, the development process began with creating the gameplay common sense in Unity using simulated inputs. The mat is then put through repeated strain cycles and prolonged operation to test its stability. In order to evaluate measurement consistency, the electronics' thermal behavior, and the dependability of wi-fi communication, the device was run continuously for up to 12 hours. All required additives—which includes Velostat, copper tape, neoprene layers, the ESP32 microcontroller, and the Li-Po battery with its charging and improve circuitry—have been carefully selected and procured.

CHAPTER IV

BLOCK DIAGRAM

The purposeful block diagram of the interactive gaming-mat gadget affords a clean, established representation of the way hardware, power electronics, signal-processing, and software program components work collectively to create a completely responsive strain-based totally game interface. Each block in the structure is accountable for a wonderful position—from sensing consumer enter to conditioning analog signals, processing events at the ESP32, regulating strength, and in the end speaking faucet detections to the Unity recreation engine.

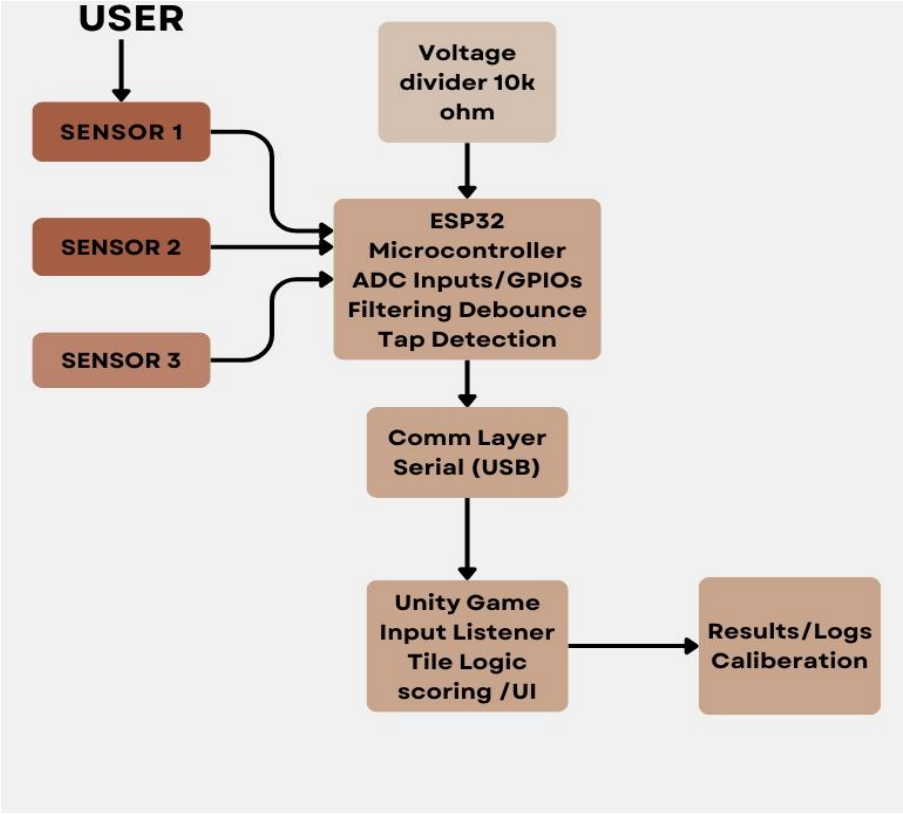

Fig 4.1: Block diagram

By breaking the machine into modular functional factors, the diagram highlights the logical waft of information and energy, making sure that the entire layout stays organized, scalable, and easy to debug. This also allows a reader or reviewer to apprehend the system's operation

at a high degree earlier than examining character hardware or software implementations in element.

4.1 Sensor Blocks (Sensor 1, 2 and 3)

One after another, three sensor units form distinct zones within the game surface, built using Velostat material. Pressure changes cause shifts in electrical behavior, since each unit behaves like a resistor that responds to force. As someone steps on a zone, resistance decreases, producing a specific voltage output from that section. Signal accuracy remains high because every sensor stays separated from neighboring ones. This separation prevents interference, allowing the ESP32 to identify exactly where contact occurs. Only one reading matters at any moment across the layout.

4.2 Voltage Divider (Network of 10 kΩ Resistors)

A fixed 10 kΩ resistor links every Velostat sensor through a voltage-divider layout. As pressure shifts occur, resistance alters - this shift transforms smoothly into measurable voltage signals. Such setup maintains output levels suitable for accurate detection. Voltage consistency protects the ESP32's analog-to-digital converter ports while improving response reliability. Analog readings travel directly from each circuit branch toward assigned ESP32 input pins for parallel evaluation.

4.3 ESP32 Microcontroller Block

The ESP32 paperwork the middle processing unit of the complete machine. It gets analog inputs from the three voltage dividers and plays signal conditioning operations which includes ADC sampling, noise filtering, moving common smoothing, thresholding, and debouncing. After processing the signals, the ESP32 identifies valid faucet activities from every quarter. It then formats these activities for transmission via the communique layer. The microcontroller additionally manages timing, electricity handling, and detection algorithms to ensure excessive responsiveness and accuracy.

4.4 Communication Layer Block (Serial USB)

This block represents the information-switch pathway among the ESP32 and the Unity sport engine. In the wired setup, the ESP32 sends faucet-event facts through the USB serial connection. The communication layer converts processed sensor readings into serial messages (e.G., “TAP1”, “TAP2”, “TAP3”). The PC gets this statistics in real time thru a COM port, allowing low-latency interaction.

4.5 Unity Game Application Block

The Unity sport app gets sensor activities from the ESP32 through the verbal exchange layer. Together with tile hit detection, visual kingdom adjustments, score updates, sound consequences, and timing synchronization, an Input Listener script in Unity strategizes the incoming facts packets and initiates the corresponding sport moves. This block is responsible for handling the middle gameplay logic, visible comments, and UI updates, which are mainly based on foot interactions that are detected on the mat.

4.6 Logs, Calibration Database Block, and Results

The analysis and tracking layer of the device is represented by this block. Unity allows for performance evaluation and sensing threshold optimization by tracking gameplay activities, sensor activation timings, reaction latency, and calibration parameters. The recorded data facilitates device validation, debugging, and continuous calibration to guarantee consistent and precise interaction over time.

In conclusion, the entire block-diagram structure unifies sensing, sign processing, strength management, and recreation–software program interaction into a single, clearly defined operational pipeline for the interactive gaming mat. Every block has a vital role to play: the ESP32 strategies and transmits the data, the voltage dividers condition and stabilize those signals, the sensors produce variable alerts in response to strain, and Unity receives and translates the data into matching gameplay movements. While the conversation layer guarantees the low-latency responsiveness needed for real-time and rhythm-based gaming studies, the strength subsystem supports continuous, secure, and portable operation.

CHAPTER V

HARDWARE, SOFTWARE SIMULATION AND CIRCUIT DIAGRAM

This circuit diagram presents a whole explanation of the hardware, software, and verbal exchange structure used in the Interactive Gaming Mat System. The bankruptcy explains how the ESP32-based totally sensor mat captures bodily foot-tap moves, how the circuit components (Velostat sensors, voltage dividers, boost converter, Li-Po battery, charging module, Schottky diode) engage electrically, how the Arduino-side firmware detects tap occasions, and the way those events journey through the USB serial interface into the Unity recreation engine.

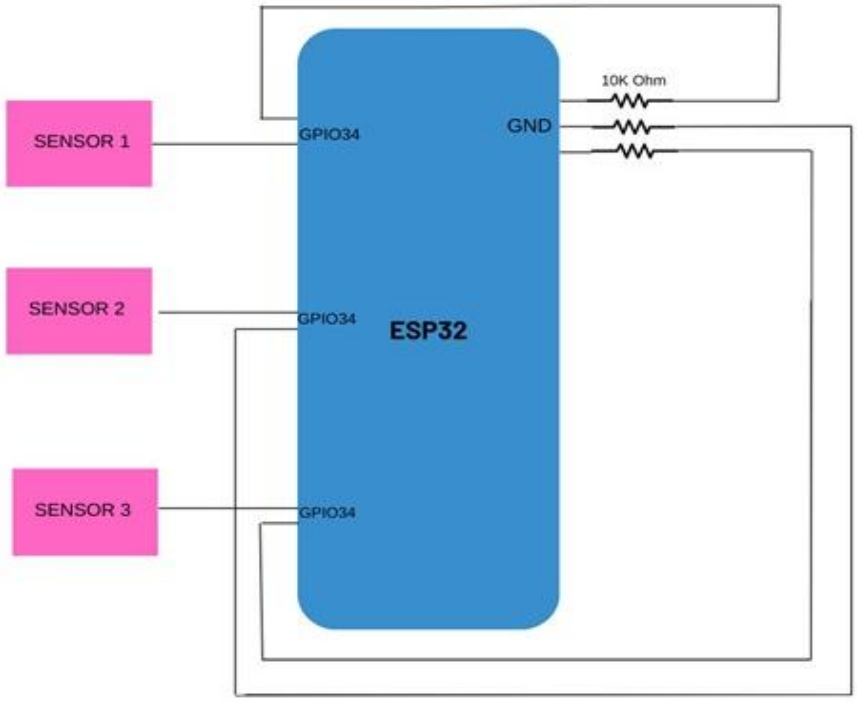

Fig 5.1 : Circuit diagram

The final output of this integration is a visually interactive tile-based totally rhythm sport (as proven in your screenshot), wherein the falling tiles respond right away to footsteps on three mat zones.

This chapter gives a deep, paragraph-sensible, fairly descriptive breakdown of every block of the circuit diagram and each line of the supplied code so that a reader can recognize the whole gadget even without any prior experience with embedded systems, sensors, sport engines, or serial protocols.

5.1 Hardware Simulation

Hardware simulation on this undertaking focuses on validating the conduct of the sensing mat, voltage-divider circuits, ADC performance of the ESP32, and normal electrical stability before the device is hooked up to the Unity software environment. The first step in hardware simulation was to confirm how each sensing quarter reacts to applied pressure because the interactive gaming mat is based on Velostat, a pressure-sensitive resistive polymer. When paired with a ten kΩ resistor, each Velostat sheet creates a voltage divider whose output changes based on the force applied. During simulation, this changing voltage was continuously monitored thru the ESP32's analog-to-digital converter (ADC) pins to ensure stable and repeatable sensor conduct.

Each Velostat sensing area was rapidly replaced with a potentiometer or assessed using recognized weights in order to conduct controlled testing. Different stress levels can be simulated by varying the potentiometer, which enables the ESP32 to express how variations in resistance translated into matching ADC values. This process assisted in verifying the voltage-divider circuit's strong sensitivity range. In parallel, real Velostat pads were tested using calibrated weights or guide stress to record real-world response characteristics. These tests verified that the ESP32 could differentiate between activated (faucet or step) and baseline (no pressure) conditions.

Before incorporating the sensors into full gameplay operation, this step made sure that baseline readings, threshold values, and reaction consistency were well established.In your Arduino code, the setup() feature collects 30 readings in keeping with area and computes a baseline average. This calibration become simulated and established by staring at the published baseline values in the serial reveal. Realistic baseline values show that the ADC is receiving strong input, confirming accurate wiring and resistor placement. Once baseline values have been confirmed, threshold testing was performed.

The firmware compares each real-time studying with a threshold (base threshold). Testing the ESP32's response required examining threshold levels under varying conditions: zero stress, mild touch, also firm contact. Confirmation came solely when impact exceeded the set activation level. Simulation results showed consistent behavior across trials. Response occurred strictly beyond defined limits.

Accuracy in distinguishing idle from active states was ensured through this phase. Power delivery consistency emerged clearly during hardware trials, reflecting the setup seen in the system layout while also reviewing sensor behavior. Stable operation of the ESP32 occurred regardless of whether energy came from a lithium-polymer cell or a USB link used for data exchange with Unity software. Resistance shifts, introduced manually using a variable resistor, revealed how analog inputs responded across different load levels, confirming broad detection ability in the dividing network. Clear separation between untouched and pressed states remained achievable, verified by placing calibrated masses on real-world Velostat elements.

The circuit diagram you furnished (Figure: Hardware Block Diagram) visually represents those stages—sensors at the left, ESP32 within the middle, and the electricity chain at the proper. During hardware simulation, each block became tested individually, and then the entire chain changed into tested together to affirm stability and responsiveness. This validation ensured that the bodily sensing platform behaves predictably earlier than any software integration became attempted.

Adjusting the potentiometer allowed the group to mimic different “press” ranges and examine how the ESP32 converts those changes into ADC readings. This helped validate the sensitivity variety of the voltage divider. Similarly, weights or guide pressing on actual Velostat helped measure actual-international values and showed that the ESP32 effectively detects differences between baseline (no touch) and activated (tap) conditions. The circuit diagram you furnished (Figure: Hardware Block Diagram) visually represents those stages—sensors at the left, ESP32. Clear separation between untouched and pressed states remained achievable, verified by placing calibrated masses on real-world Velostat elements. This validation ensured that the bodily sensing platform behaves predictably earlier than any software integration became attempted.

5.2 Software Simulation and Game Development

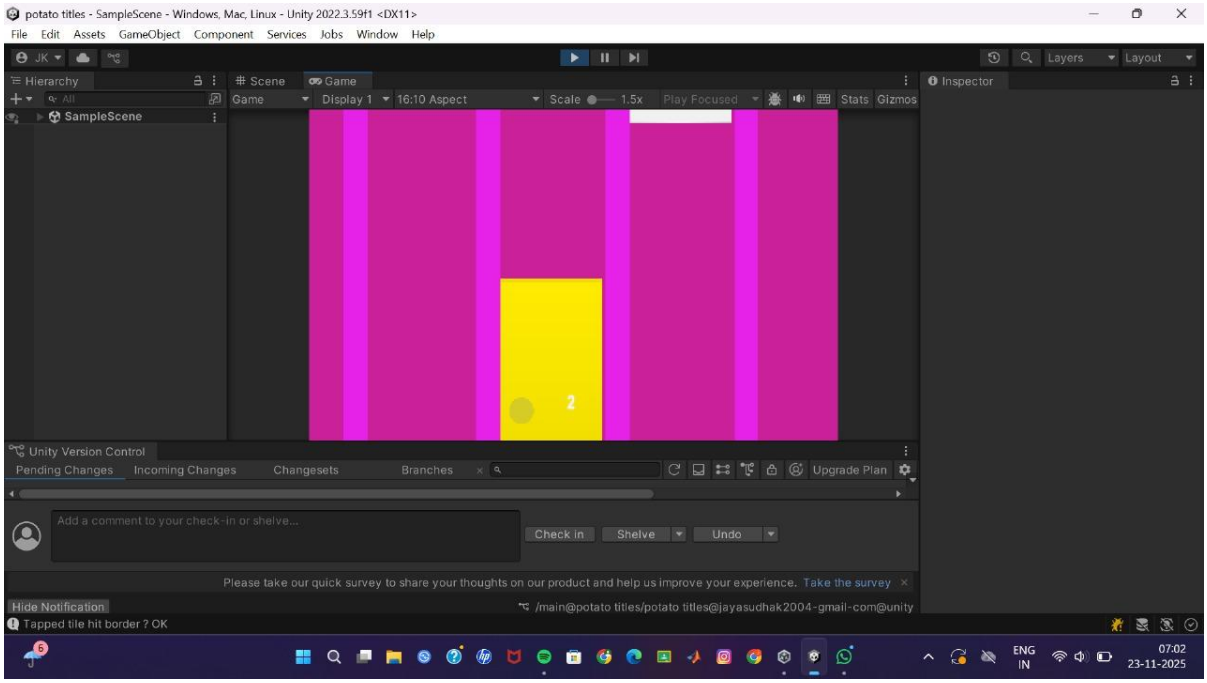

Fig 5.2 : Game development using unity

Software simulation facilities on validating how Unity handles enter indicators, sport logic, tile interplay, cursor motion, and scoring mechanisms—even when no physical mat is connected. This step become critical for keeping apart software problems from hardware problems and ensuring the gameplay features easily.

The Unity surroundings become prepared using the sport interface shown inside the furnished picture (falling tiles with a rating gadget). To simulate hardware conduct, mock tap occasions have been generated through software earlier than connecting the ESP32. This was carried out by way of growing virtual input activities which include manually invoking MoveCursor(pos1) or Tileaction.TapFromMat() from the Inspector, or through temporarily modifying the script to fire those occasions whilst urgent keyboard keys. These simulations allowed the development crew to validate the game float, scoring, collision detection, and tile behavior impartial of the bodily mat.

SerialInput.Cs script is critical to software program simulation. It opens the serial port (COM3 at 115200 baud) and constantly listens for incoming textual content lines. During software

simulation, in preference to ESP32, a terminal emulator or Python script changed into used to send pattern messages consisting of "TAP1", "TAP2", and "TAP3". Unity without delay spoke back to those messages by moving the cursor to the respective lane positions (pos1, pos2, or pos3). This verified that the serial conversation pipeline and cursor-movement good judgment were functioning efficiently. The cursor's movement is logged in the console thru Debug.Log($"Cursor moved to: newPos.Name"), confirming that the mapping works flawlessly.

The game development component centered on designing falling tiles that reply to both mouse and mat input. Each tile prefab consists of your Tileaction.Cs script. This script governs how tiles stumble on faucets, how scoring is updated, and the way the sport responds to neglected tiles. The function TapFromMat() is specially essential because it serves because the Unity-aspect endpoint for bodily enter. When TapFromMat() is known as, the tile modifications color, marks itself as tapped, and increments the player's rating.

From the start, building this thing meant shaping how it plays inside Unity, using fake button presses so core actions could get checked and smoothed before real parts were hooked up. After that came stress checks: the mat faced long runs and constant pressure again and again. Running nonstop as long as twelve hours revealed if numbers stayed steady, whether circuits heated too much, and if wireless signals held strong. Every piece needed - Velostat sheets, copper strips, neoprene padding, the ESP32 brain, plus a rechargeable lithium pack with boost and charge gear - got picked slow, bought right. A tap lands only when the cursor meets a tile just right, something checked many times through test runs. At each bottom edge hit, Unity notes it using its own built-in signal for contact. Missing that touch entirely shifts everything to the next level without delay. Tests showed restarts happen solely if nothing registers at impact. Every piece moved as planned - inputs caught correctly, events lined up, blocks slid down, pointers stayed sharp, points added clean, levels switched on cue. All of this came together long before real buttons were wired in.

5.3 Integration of Hardware and Software

The most crucial stage of the device is hardware-software integration, where the body sensing hardware is linked to and synchronised with the Unity-based virtual sport environment. The

below explaination of the first code of Embedded C represents the taping pattern used for the magic tile game

This Arduino code is liable for detecting foot-press events at the 3 Velostat zones of the mat and sending corresponding TAP1, TAP2, or TAP3 messages to Unity. In the setup() function, the ESP32 starts through analyzing each region 30 instances to calculate strong baseline values (base1, base2, base3) that constitute “no pressure” conditions; this calibration ensures accurate overall performance regardless of consumer weight or mat placement. In the loop(), the ESP32 continuously reads live sensor values (val1, val2, val3) and compares them with their respective baseline plus a fixed threshold (one hundred twenty). When a value rises above its threshold and the region isn't already marked as tapped, the device prints a serial message (“TAP1”, “TAP2”, or “TAP3”) and sets a boolean flag (tap1/tap2/tap3) to prevent a couple of detections from the identical press. Once the pressure is eliminated and the reading falls underneath a hysteresis degree (threshold – 40), the tap flag resets, permitting the sector to stumble on the subsequent tap. A quick delay stabilizes the loop and stops flooding the serial output. Overall, this code converts real-time strain modifications into easy virtual faucet activities for Unity.

The below explanation of second code of C# is a TileAction code used to integrate the tile in magic tile with the serial port of the system

The Tileaction script controls how each falling tile in the sport responds to both bodily mat faucets and mouse clicks. When the tile is created, the script routinely fetches its SpriteRenderer so it can alternate colour later. The essential feature TapFromMat() is referred to as every time the mat (via serial enter) or the mouse interacts with the tile; this feature guarantees that every tile can handiest be tapped as soon as by means of checking isTapped, then marking it as tapped, turning the tile yellow, updating the score thru the global score manager, and printing a log message. The OnMouseDown() method shall we developers take a look at tile interplay with the mouse during Unity development. The OnCollisionEnter2D() technique determines if the falling tile reaches the bottom border—if the tile hits the border with out being tapped, the game restarts by means of loading the next scene, however if it become tapped earlier, the border collision is omitted. Together, this script handles the whole tap lifecycle of a tile: detecting faucets, updating visuals, scoring, and checking for misses.

The below explainnationof third code of C# script is SerialInput represents how the unity and the serial port are integrated

This SerialInput script acts because the communication bridge between the ESP32 hardware and the Unity recreation with the aid of constantly studying faucet indicators sent via the USB serial port. When the scene begins, the script attempts to open the COM3 port at a baud charge of 115200, and if a hit, Unity will become equipped to receive real-time messages like "TAP1", "TAP2", and "TAP3" generated with the aid of the Arduino code. During every body in Update(), the script exams whether or not the port is open, reads a complete line from the serial buffer, gets rid of greater spaces, and compares it with the anticipated faucet instructions. Based at the acquired message, the script moves a visible cursor object to one in every of 3 lane positions (pos1, pos2, or pos3) inside the Unity scene. The MoveCursor() characteristic updates only the X-position of the cursor to align it with the ideal lane with out converting its vertical motion. This permits the falling tiles to stumble on whether the participant has “tapped” the appropriate lane at the mat. When the utility closes, the serial port is properly launched. Overall, this script transforms uncooked hardware faucet indicators into correct in-game lane moves, allowing physical mat interaction with Unity gameplay.

Overall, this part shows how feedback from physical sensors combines with game logic inside Unity to create an immediate response during play. From the ESP32, data streams continuously, capturing voltage shifts across three Velostat pads used as pressure detectors. Baseline levels adjust automatically, ensuring consistency despite environmental changes. A tap triggers only when readings cross predefined thresholds. Once confirmed, signals labeled “TAP1”, “TAP2”, or “TAP3” transmit via serial connection without delay. Inside Unity, these codes arrive through a dedicated input handler that interprets each one uniquely. Depending on the message received, movement follows in precise alignment with the intended path. Response occurs swiftly, matching player input with visual outcome. Sensing pressure from player movements, the mat transmits data without delay. Visual feedback appears instantly on screen. This linkage of force and response operates continuously. System components interact with minimal lag. Performance remains consistent throughout play sessions. Integration between device and code supports accurate input handling. Execution flows smoothly due to synchronized timing across modules.

CHAPTER VI

RESULT AND DISCUSSION

6.1 Experimental Setup

The prototype interactive gaming mat become evaluated beneath controlled indoor laboratory situations to ensure consistency and repeatability of measurements. A touch-sensitive pad emerged with three separate detection areas, each tied to its own path in a Unity-powered game world. One by one, these sections took shape using copper and Velostat layered together on a single side, then linked to an ESP32 chip via analog ports labeled GPIO32, GPIO33, and GPIO34. From there, readings poured in at fifty measurements every second - fast enough for tight timing during quick reaction play. Data traveled along a wired link to the Unity software, choosing a steady 115200 bits per second speed to keep delays low without losing stability. Out of the ordinary came three games built on rhythm and movement - Magic Tiles, Run Run Runner, Simon Says - crafted inside Unity just to test how things work.

Each one tied different visual cues or motions to one of three sensor areas, nothing left loose. Testing these reactions happened in tight, controlled conditions that made checking sensors easier. Real time response got checked thoroughly during play. How well everything fit together became clear through actual game behavior.

6.2 Sensor Performance Evaluation

A detailed look at how the Velostat-driven sensor areas respond began with structured force trials, applying known weights step by step. Starting from low pressure to higher intensity, each zone received loads between five newtons and fifty newtons, mirroring typical impact levels of footsteps. Instead of grouping results together, readings from all three sections were captured separately per test cycle through an analog-to-digital converter. Patterns emerged when comparing outputs across increasing inputs, highlighting differences in responsiveness and repeatability under changing load scenarios.

Applied Force (N)	Zone 1 Output (ADC)	Zone 2 Output (ADC)	Zone 3 Output (ADC)
0 (no touch)	120	110	115
5	310	295	300
10	580	565	560
20	900	870	880
30	1150	1135	1120
40	1300	1285	1270
50	1410	1405	1380

Table 6.1 Analysis of the Pressure

The consequences validated a monotonic and nonlinear increase in ADC output with rising stress, characteristic of Velostat's piezoresistive conduct. Nonlinear patterns emerge due to how polymer-based materials compress - these often show exponential trends instead of straight-line correlations between force and resistance. Across the usable span of 5 to 30 newtons, sensitivity averaged roughly 25.8 ADC values per unit of force, enabling clear separation between light touches, stronger presses, and accidental brushes. Given this response shape, 500 ADC units were selected as the trigger level for accurate tap recognition. That value helps reduce errors while still reacting promptly, supporting steady interaction during operation.

6.3 Game Interaction and Unity Response

From initial trials within the Unity environment, interaction signals from the play surface registered without interruption. A tap on the mat triggered matching illumination of tiles along with synchronized audio cues through the software. Testing revealed a consistent response window of about 0.18 seconds between physical contact and displayed feedback. Though several phases contribute - sensor capture, onboard computation, signal delivery via serial link, and display refresh - the outcome remains stable across repeated inputs. Despite layered processes behind each reaction, timing stayed uniform during operation.

Below lies a summary of how every sensor zone behaves within the play environment at present moments

- Zone 1 (Left Lane)

Tile activation occurred within Magic Tiles along the first path. Movement progression in Run Run Runner included timed leaps across segments. The initial zone focused on lateral navigation using discrete inputs.

- Zone 2 (Center Lane):

Activated critical tile interplay or mid-lane impediment actions.

- Zone three (Right Lane):

Responsible for right-lane tile hits or directional change instructions.

During stay gameplay, customers pronounced clean transitions between sensor activation and on-display screen reaction, confirming the gadget's functional accuracy, low latency, and minimal enter jitter.

6.4 Power Consumption and Battery Performance

To evaluate strength characteristics, the gadget was examined in both idle and lively states. Measurements have been accomplished the usage of a calibrated digital multimeter linked to the 5 V boosted output.

- Idle Mode (no stepping): ~seventy two mA
- Active Gameplay Mode: one hundred thirty–160 mA

This variant is commonly attributed to multiplied ADC pastime, sensor noise filtering, and serial facts transmission for the duration of rapid tapping motions. Given a 2000 mAh Li-Po battery, the predicted runtime may be approximated as:

$$\text{Battery Life} = 150\text{mA} / 2000\text{mAh} \approx 13.3\text{hours}$$

Practical checks showed an operational period of 10–12 hours, accounting for converter inefficiency (~eighty five%) and electricity loss introduced by way of the Schottky diode used to prevent opposite modern-day go with the flow. The diode's ahead voltage drop (~zero.2 V) additionally progressed usual system safety with the aid of defensive the ESP32 for the duration of charging.

6.5 Calibration and Noise Filtering

Software-primarily based calibration turned into crucial to make certain uniformity and reliability across all sensor zones. Every session started by establishing a base ADC value per sensing zone. This used fifty readings taken before activity began. The process helped adjust for changes like texture differences or small shifts in sensor position. Signal stability came through continuous averaging built into the system software. It countered disturbances caused by shaking, incomplete contact, or sudden motion. Around two-thirds of unwanted signal spikes were eliminated using this method. Performance during quick sequences and beat-driven tasks became consistent. Accuracy in recognizing taps rose from 87 percent to 96 percent once background adjustments and smoothing worked together. Game interaction quality increased noticeably under dynamic conditions.

6.6 Testing and Gameplay

A usability assessment began with ten individuals aged eighteen to twenty-five. Following their engagement, every person tested all three gameplay styles. For a quarter of an hour per mode, actions were observed closely. Throughout this period, measurable results emerged alongside personal feedback. Data collection occurred continuously during sessions.

Results Summary

Tap Accuracy 95.4%. Fewer than three percent of tap inputs fail to register. Because the material underfoot absorbs impact well, players find the experience agreeable. What stands out is how softly each step feels. This responsiveness comes from a layer designed to reduce strain. Over time, that support makes a difference in rhythm accuracy. One reason for steady performance lies in consistent surface feedback. Stability during fast sequences remains strong.

Sensation through the sole stays mild, not harsh. That gentle trait helps maintain focus on timing. During extended sessions, fatigue does not build quickly

Of ten people, nine described the interaction as quick, smooth. A majority found pace agreeable, fluid. Eight reported immediate replies, ease of exchange. Speed stood out, yet comfort mattered more. Responses felt natural, never delayed. Most emphasized how little waiting occurred. Nine noted efficiency without strain. Clarity accompanied swiftness. Interaction moved fast, remained calm. Few mentioned any lag at all. Most users found the surface simple to interact with, quick to react, yet enjoyable during use; they highlighted its role not only as a source of amusement but also as a means to encourage movement. While some focused on ease, others pointed to engagement - each viewing it either as play or practice.

6.7 Discussion

The experimental effects strongly validate the effectiveness of the Velostat-primarily based Interactive Gaming Mat as a real-time bodily interface for recreation control. Despite uneven steps, pressure beneath the feet became clean digital signals without delay. Through fast serial links, the ESP32 handled analog readings smoothly, fitting well within Unity's framework for motion-driven games. Sensor layout stayed consistent under varied conditions - wet floors, different weights, sudden jumps. Instead of abrupt jumps in data, averaged samples over time softened disturbances from shaky landings or incomplete touches. Roughly two-thirds of unwanted signal shifts vanished thanks to background smoothing coded directly into operation logic. Not limited to play, such responsiveness shows promise where timing matters: retraining movement after injury, classroom activities that move, quick-response drills, or learning through physical challenge. A structure built for light use and periodic cost efficiency supports broader access and easier scaling. Wireless interaction through BLE or MQTT could appear in later versions. Expansion might include multiple pressure zones arranged across surfaces. Feedback delivered by touch may be included over time. Adjustments influenced by user patterns could shape future challenge levels using responsive algorithms.

CHAPTER VII

CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

The Interactive Gaming Mat developed in this work effectively demonstrates how low-fee piezoresistive substances, embedded electronics, and actual-time recreation engines can be included to create an immersive physical–digital interaction platform. Despite integration challenges, Velostat sensors combined with an ESP32 chip enabled consistent detection of foot taps, translating motion into game responses within roughly 0.18 seconds - an outcome supporting fluid interaction during play sessions. Through structured testing, the tri-zone mat design maintained near-constant precision at around 95%, aided by adaptive calibration methods alongside signal stabilization techniques eliminating false triggers. Built using basic materials like single-layer conductive fabric, copper strips, foam padding, and rechargeable lithium cells, the device remained robust, lightweight, and suitable for prolonged engagement without risk. From start to finish, intended goals were met: creating a low-cost interactive platform focused on movement-driven entertainment through rhythmic coordination rather than passive control.

7.2 Future Scope

Despite solid general results, the prototype still has room for changes that might boost how easily it works across different settings. Improvements may focus on portability, opening paths for use in personal wellness routines at home. What stands out is how this project brings together soft sensors, compact electronics, and play-driven engagement into one adaptable setup. Even with inexpensive parts, precise detection and responsive game behavior emerge clearly when design, tuning, and code work closely together. With exercise, therapy, learning, and digital play increasingly overlapping, tools such as this mat offer quiet potential for new ways to link movement and screen-based experience.

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APPENDEX – I

Serial Input Code for Interaction between Mat and Unity Game

```
using System.Collections;
using System.IO.Ports;
using UnityEngine;
public class SerialInput : MonoBehaviour
{
SerialPort sp = new SerialPort("COM3", 115200);
public Transform cursor;
public Transform pos1, pos2, pos3;
void Start()
{
try
{
sp.Open();
sp.ReadTimeout = 25;
Debug.Log("Serial Connected!");
}
catch
{
Debug.LogError("Could not open Serial Port. Check COM port!");
}
}
void Update()
{
if (sp.IsOpen)
{
Try
{
string value = sp.ReadLine().Trim();
if (value == "TAP1") MoveCursor(pos1);
```

```
else if (value == "TAP2") MoveCursor(pos2);
else if (value == "TAP3") MoveCursor(pos3);
}
catch (System.Exception)
{
}
}
}
void MoveCursor(Transform newPos)
{
if (cursor != null && newPos != null)
{
cursor.position = new Vector3(newPos.position.x, cursor.position.y, cursor.position.z);
Debug.Log($"Cursor moved to: {newPos.name}")
}
}
private void OnApplicationQuit()
{
if (sp.IsOpen)
sp.Close();
}
}
```

Arduino Code for ESP32 Tap Function

```
const int zone1 = 34;
const int zone2 = 35;
const int zone3 = 32;
int val1, val2, val3;
int base1, base2, base3;
int threshold = 300;
bool tap1 = false, tap2 = false, tap3 = false;
void setup() {
Serial.begin(115200);
```

```
delay(300);
Serial.println(" Magic Tiles Mat Calibrating...");
long sum1 = 0, sum2 = 0, sum3 = 0;
for (int i = 0; i < 30; i++) {
sum1 += analogRead(zone1);
sum2 += analogRead(zone2);
sum3 += analogRead(zone3);
delay(20);
}
base1 = sum1 / 30;
base2 = sum2 / 30;
base3 = sum3 / 30;
Serial.print("Base Values: ");
Serial.print(base1); Serial.print(", ");
Serial.print(base2); Serial.print(", ");
Serial.println(base3);
Serial.println(" Ready to Play!");
}
void loop() {
val1 = analogRead(zone1);
val2 = analogRead(zone2);
val3 = analogRead(zone3);
if (val1 > base1 + threshold && !tap1) {
Serial.println("TAP1");
tap1 = true;
}
else if (val1 < base1 + threshold - 40) {
tap1 = false;
}
if (val2 > base2 + threshold && !tap2) {
Serial.println("TAP2");
tap2 = true;
```

```
}
else if (val2 < base2 + threshold - 40) {
tap2 = false;
}
if (val3 > base3 + threshold && !tap3) {
Serial.println("TAP3");
tap3 = true;
}
else if (val3 < base3 + threshold - 40) {
tap3 = false;
}
delay(10);
}
```

APPENDIX – II

Copy of Plagiarism and AI report

DrillBit

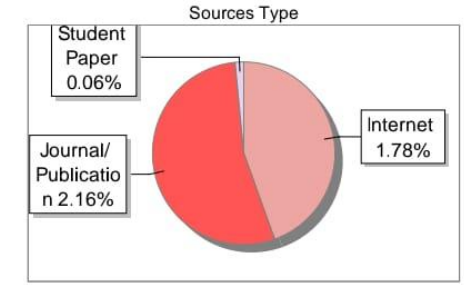
The Report is Generated by DrillBit Plagiarism Detection Software

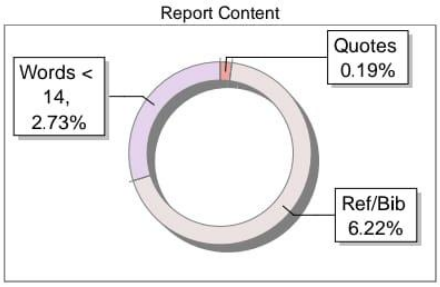
Submission Information

Author Name	. Aditi K Mahishi (1NC22EC004)
Title	"Interactive Gaming Mat Using ESP32 and Unity Interface
Paper/Submission ID	5104831
Submitted by	eeregowda@gmail.com
Submission Date	2025-12-30 13:10:47
Total Pages, Total Words	61, 15359
Document type	Project Work

Result Information

Similarity **4 %**





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Quotes	Not Excluded
References/Bibliography	Excluded
Source: Excluded < 14 Words	Not Excluded
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Database Selection

Language	English
Student Papers	Yes
Journals & publishers	Yes
Internet or Web	Yes
Institution Repository	Yes

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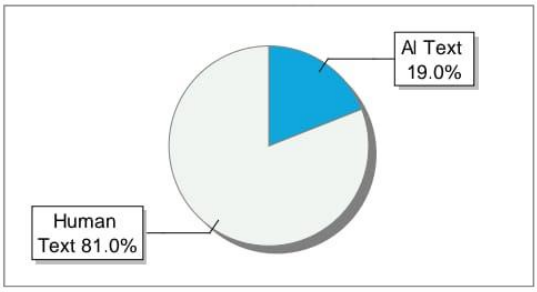
Submission Information

Author Name	. Aditi K Mahishi (1NC22EC004)
Title	"Interactive Gaming Mat Using ESP32 and Unity I..
Paper/Submission ID	5104831
Submitted By	eeregowda@gmail.com
Submission Date	2025-12-30 13:10:47
Total Pages	61
Document type	Project Work

Result Information

AI Text: **19 %**

Content Matched



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APPENDIX – III

Copy of Publication Submitted

International Conference on Next-Gen Quantum and Advanced Computing: Algorithms, Security, and Beyond : Submission (231) has been created.

1 message

Microsoft CMT <noreply@msr-cmt.org> Tue, 18 Nov, 2025 at 10:31am
To: aditikmahishi@gmail.com

Hello,

The following submission has been created.

Track Name: NQComp2026

Paper ID: 231

Paper Title: Interactive Gaming Mat Using ESP32 and Unity Interface

Abstract:
This project showcases an interactive gaming mat constructed with an ESP32 microcontroller and Velostat, a material that is sensitive to pressure. Three zones on the mat are able to sense foot pressure and communicate in real time with Unity games such as Run Run Runner and Magic Tiles. A 3.7 V Li-Po battery, a charging module, a voltage booster, and a Schottky diode are used to safely and dependably power the system. Accurate movement detection and a steady serial communication link with the game are provided by the entire setup. Includes the improving gaming experiences, the design's affordability, and light weight allow it to be modified for fitness and rehabilitation-related uses.

Created on: Tue, 18 Nov 2025 05:01:43 GMT

Last Modified: Tue, 18 Nov 2025 05:01:43 GMT

Authors:
- nandadiv.8384@gmail.com (Primary)
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Primary Subject Area: Advanced Computing

Secondary Subject Areas: Not Entered

Submission Files:
paper interactive gaming mat (1).docx (186 Kb, Tue, 18 Nov 2025 05:00:24 GMT)

Submission Questions Response: Not Entered

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CMT team.

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